

# Jack Levitt

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Project Portfolio: [tinyurl.com/levittfolio](https://tinyurl.com/levittfolio)

## EDUCATION

### Princeton University

Expected May 2029

*Bachelor of Science in Engineering in Mechanical and Aerospace Engineering; GPA: 3.9/4.0*

**Relevant Coursework:** Small Spaceflight Hardware Projects, Engineering Design, Thermodynamics, Solid Mechanics

## RELEVANT EXPERIENCE

### Princeton Rocketry IREC Team Lead

September 2025 - Present

- Led a cross-functional team of 50+ engineers designing, building, and testing a high-power rocket targeting 30,000 ft apogee and Mach 2+ for the Intercollegiate Rocket Engineering Competition, overseeing propulsion, avionics, recovery, payload, and structural subsystem integration.
- Led recovery systems engineering for the rocket, developing parachute deployment mechanisms and conducting ground testing to validate mission-critical flight hardware.
- Conducted CFD simulations of rocket geometries to evaluate aerodynamic stability and drag performance, and designed/fabricated structural bulkheads using CAD and CNC manufacturing processes.

### Reworld Mechanical/Operations Engineering Intern

May 2026 - August 2026

- Performed and authored plant-wide structural analysis across 50+ mechanical systems, evaluating load paths, support conditions, and equipment integrity, flagging 3 critical systems for engineering review and preventing an estimated 7 days of downtime over the next year.
- Developed and revised P&IDs and CAD models for plant equipment and utility systems, producing accurate engineering drawings that supported communications between operations, maintenance, and external vendors.
- Developed technical engineering documentation detailing component mechanics, failure modes, and maintenance principles of critical plant equipment to improve operator understanding of system functionality.

### Orbis-Metallicum LLC Manufacturing Systems Engineering Intern

December 2025 - Present

- Built and optimized a global manufacturing database of EV gigacasting, integrating casting architecture, machine tonnage, and component-level design specifications across 12 OEMs and 40+ facilities, comprising 2,000+ data points.
- Evaluated graphene and graphene-doped aluminum/copper composites as next-generation electrical conductors, benchmarking theoretical thermal and electrical conductivity gains against incumbent materials to assess feasibility for weight-sensitive applications in aerospace and high-power electronics.
- Developed first-principles models of automotive battery chemistries, analyzing material composition, manufacturing processes, and electrochemical performance tradeoffs to evaluate future technology adoption.

### Princeton Student Ventures Director of Events

September 2025 - Present

- Built a network of 150+ technical founders, VCs, and industry leaders across aerospace, robotics, and advanced manufacturing to source and evaluate early-stage deep-tech innovations.
- Conducted technical due diligence on emerging startups, assessing engineering feasibility, IP defensibility, and manufacturing scalability to inform investment recommendations.
- Authored technical memos evaluating product architecture, engineering risk, and go-to-market pathways for 15+ startups, directly informing investment decisions on over \$50,000 in capital.

## RESEARCH EXPERIENCE

### Princeton University Rabitz Lab Undergraduate Researcher

August 2026 - Present

- Collected and analyzed 100+ cryogenic infrared spectra of biological small molecules to generate high-quality reference datasets, directly supporting development of a novel microscopy instrument designed to map the spatial distribution of molecules within cells in their native state.
- Performed spectroscopy experiments under cryogenic conditions, applying data processing and spectral analysis techniques to characterize biological molecules while ensuring reproducible, high-fidelity experimental results.
- Collaborated with faculty and research mentors to interpret infrared spectroscopy data and validate reference spectra, accelerating efforts to develop new methods for studying cellular metabolism and molecular organization.

## TECHNICAL SKILLS

**Software:** MATLAB, Python, Java, C++, SolidWorks, PTC Creo, Autodesk Fusion 360, AutoCAD, OpenRocket, LaTeX

**Analysis & Design:** Computational Fluid Dynamics, Structural Analysis, Manufacturing Design, Systems Engineering

**Other:** Microsoft Suite, Spanish (Intermediate), NYS Seal of Biliteracy